

PAYMENT RAIL TAXONOMY AUDIT

PAYMENT RAIL TAXONOMY

FAMILY R-001: GENAI-ENHANCED PAYMENT NETWORK MESSAGE FORGERY

A. IDENTITY

Field	Value
Attack ID	R-001
Attack Family	GenAI-Enhanced Payment Network Message Forgery
Attack Subtypes / Variants	AI-generated SWIFT MT/MX message injection; Hyper-personalized phishing for payment operator credentials; Deepfake voice verification bypass; AI-generated bank correspondence for fraudulent payment instructions; Compromised operator credential abuse with AI-generated messages
Primary Lifecycle Stage	Payment Rail (Stage 9)
Secondary / Cross-stage Stages	Authentication (Stage 4 – credential theft); Payment Initiation (Stage 5 – instruction creation); Settlement (Stage 13 – final fund movement)
Target / Actor	SWIFT network messaging infrastructure; payment operators; SWIFT participant authentication; cross-border payment routing; settlement messaging
Attack Objective	Inject fraudulent cross-border payment instructions into the SWIFT network to move funds from victim banks to attacker-controlled accounts, bypassing participant authentication and message integrity controls
Network Control Targeted	Participant Authentication; Message Integrity; Payment Operator Trust; Settlement Messaging

B. FRAUD UNDERSTANDING

What is the attack?

The attacker compromises a financial institution's SWIFT environment (through AI-enhanced phishing, credential theft, or insider access) and sends fraudulent SWIFT MT (e.g., MT103, MT202, MT760) or MX messages to move funds to attacker-controlled accounts. GenAI amplifies this through hyper-personalized phishing for credential harvesting, deepfake voice confirmation to bypass callback verification, and AI-generated payment messages that appear indistinguishable from legitimate instructions.

The attack targets the **SWIFT network's trust model**—the network assumes messages from authenticated participants are legitimate. Compromised credentials violate this trust.

Traditional mechanism

Before GenAI, SWIFT message forgery worked through:

- **Compromised bank environments:** Attackers infiltrated a bank's internal network and stole SWIFT credentials
- **Manual message crafting:** Attackers manually crafted fraudulent SWIFT MT messages using knowledge of SWIFT formats
- **Credential theft:** Payment operator credentials were stolen through phishing, insider complicity, or malware
- **Timing exploitation:** Messages were sent outside normal working hours or over holidays to delay detection

The Bangladesh Bank attack (2016) exemplified this—attackers sent 35 fraudulent SWIFT messages attempting to transfer nearly \$1 billion, with 5 messages successfully executing approximately \$101 million.

GenAI transformation

Dimension	Pre-GenAI	GenAI-Transformed
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Dimension	Pre-GenAI	GenAI-Transformed
Credential theft	Generic phishing with grammar errors	Hyper-personalized LLM-generated phishing referencing real projects, SWIFT codes, colleagues, and recent transactions
Voice verification	Human impersonation (poor quality, limited)	Deepfake voice cloning from 3-second samples; real-time voice conversion
Message generation	Manual crafting, template-based errors	LLM-generated SWIFT messages with perfect format, correct BIC codes, contextually accurate references
Reconnaissance	Manual OSINT; internal knowledge	AI-powered analysis of transaction timing and sizes to identify optimal insertion moments
Bank correspondence	Manual forgery	AI-generated professional-grade compliance documents, verification letters, and regulatory filings
Scale	Opportunistic, manual	Industrial-scale, automated campaigns

Why GenAI is load-bearing

If GenAI is removed, the attacker loses the ability to:

- 1. Generate grammatically perfect, contextually accurate payment redirection emails at scale
- 2. Create deepfake voice clones to bypass callback verification
- 3. Hyper-personalize attacks based on OSINT of individual operators
- 4. Scale credential theft from manual to industrial campaigns
- 5. Generate professional-grade bank correspondence without specialized skills

However, the core attack mechanism—compromising bank environments and sending fraudulent SWIFT messages—exists without GenAI. GenAI **amplifies** rather than creates the attack.

HARD GenAI Removal Test Result

Component	Result	Rationale
Credential theft via phishing	PASS	GenAI enables hyper-personalized phishing at scale—"what once required deep technical skill can now be executed with a prompt"
Message crafting	FAIL	SWIFT message formats are structured and documented. GenAI automates but does not transform
Voice verification bypass	PASS	Deepfake voice cloning enables convincing impersonation that did not exist before GenAI
Reconnaissance & timing	PARTIAL	GenAI accelerates but does not create the capability

Overall: PARTIAL — GenAI significantly amplifies the attack but does not fundamentally create it.

Prerequisites

- Access to a bank's internal SWIFT environment (via compromised credentials or insider access)
- Knowledge of SWIFT message formats (MT/MX) and institutional BIC codes
- Ability to submit messages to the SWIFT network through the bank's interface
- (For GenAI-enhanced variants) Access to LLMs for phishing generation; voice samples for cloning (3 seconds sufficient)

Attack Flow

- 1. Attacker conducts reconnaissance on target bank (OSINT, AI-powered analysis)
- 2. Attacker uses GenAI to generate hyper-personalized phishing emails targeting SWIFT operators
- 3. Operator credentials compromised via phishing (or via insider access)
- 4. Attacker uses GenAI to craft fraudulent SWIFT payment messages (MT103, MT202, MT760)
- 5. Attacker logs into compromised SWIFT environment (Alliance Access/SWIFTNet)
- 6. Attacker submits fraudulent payment instruction during normal business hours (to avoid timing flags)
- 7. SWIFT network processes message as legitimate (authenticated participant)
- 8. Message routed to receiving financial institution
- 9. Receiving institution processes payment; funds credited to attacker-controlled account
- 10. Funds moved through mule accounts → crypto → cash-out
- 11. Detection occurs days later during reconciliation (if at all)

Payment Rail Lifecycle Point

Fraud enters at **Network Submission** — the fraudulent message is submitted from an authenticated participant endpoint and the network processes it in good faith.

Lifecycle Stage	Role in Attack
Instruction Creation	Attacker generates fraudulent MT/MX message using internal SWIFT workstation
Network Submission	Message submitted to SWIFT network from compromised institution's Alliance Access
Network Processing	SWIFT validates message format, BIC routing, participant status
Routing	Message routed to receiving institution via correspondent network
Settlement Handoff	Settlement instructions executed; funds moved irrevocably

Participant Role

Participant	Role
Sending Institution (Compromised)	The attacker's entry point. Credentials are stolen; environment is used to submit fraudulent messages
SWIFT Network	The victim of trust abuse. Processes fraudulent message as legitimate
Receiving Institution	Processes fraudulent instruction in good faith; credits attacker-controlled account
Attacker	Compromises credentials, crafts messages, manages mule accounts

Downstream Payment / Financial Consequence

- Direct financial loss to sending institution (millions of dollars)
- Attacker funds moved through layering → crypto → cash-out
- Regulatory penalties for sending institution
- Reputational damage
- Systemic risk to cross-border payment infrastructure
- Customer/victim institutions suffer losses

C. DETECTION INTELLIGENCE

Observable Signals

Signal Category	Specific Signals	Why It Matters	Detection Method	Network-Wide Intelligence Required?
Transaction	Amount deviation from historical pattern; urgent payment flags; narrative field anomalies	Indicates potential fraudulent instruction	Rules; ML anomaly detection; LLM-based NLP	No
Transaction	New or unusual beneficiary BIC/account; beneficiary in high-risk jurisdiction	Mule account indicator	Entity resolution; threat intelligence; ML risk scoring	Yes
Network	Unusual message volume from specific participant	Compromised institution indicator	Network-level analytics	Yes
Network	Unusual routing paths; multiple intermediary correspondents	Evasion of direct relationships	Graph analytics	Yes
Participant	Operator credential usage at unusual times; login from unusual location	Credential compromise indicator	Rules; temporal anomaly detection	No
Participant	Rapid sequence of messages from same operator	Automation/bot behavior	Velocity rules; anomaly detection	No
Routing	Unusual BIC combinations; routing through non-preferred correspondents	Routing manipulation or unauthorized instruction	Rules; supervised ML	Yes
Temporal	Off-hours or holiday submissions; rapid succession of large payments	Timing evasion	Rules; temporal anomaly	No
Behavioral	Unnatural navigation in SWIFT interface; superhuman speed	Bot/automation indicator	Behavioral biometrics	No
Graph/Network	Beneficiary connections to known mule/crypto networks; rapid onward movement	Money laundering indicator	Graph Neural Network; Entity resolution	Yes – Essential
Message/State	SWIFT message format anomalies; narrative field unnatural phrasing	AI-generated content indicator	NLP; LLM-based detection (perplexity, stylometry)	No

Signal Category	Specific Signals	Why It Matters	Detection Method	Network-Wide Intelligence Required?
Message/State	Missing or inconsistent regulatory references	Incomplete/inconsistent instruction	Rules	No
Geographic	Payment to jurisdiction with no prior relationship; unusual corridor	Red flag for trade-based laundering	Rules; anomaly detection	Yes
Cross-Rail/Participant	Rapid crypto conversion following receipt	Mule/cash-out pattern	Graph analytics; threat intelligence	Yes – Essential

Evasion Considerations (Defensive Threat Modeling)

An adaptive attacker might attempt to:

Attack Dimension	Evasion Tactic	Why Static Controls Fail	Adaptive Monitoring Should Observe
Timing	Submit messages during normal business hours; avoid weekends/holidays	Time-based rules are easily bypassed	Subtle deviations from historical timing distribution
Amount	Match historical transaction sizes; avoid large round numbers	Amount thresholds are easily bypassed	Deviations from historical amount distribution
Beneficiary	Pre-establish beneficiary accounts weeks/months in advance; use legitimate-looking shell companies	New beneficiary rules fail	Graph anomalies (beneficiary connected to mule networks)
Operator Behavior	Use compromised operator credentials during scheduled shifts; mimic human typing	Behavioral biometrics may detect if mimicry is poor	Subtle behavioral anomalies
Phishing Quality	Use AI-generated phishing that perfectly matches internal tone	Traditional email filters fail	NLP-based detection; AI-generated content detectors
Voice Verification	Use high-quality deepfake voice with realistic background noise	Human verification fails	Multi-factor verification; biometric liveness
Message Generation	Generate messages matching institution-specific formatting	Format validation fails	Metadata analysis; NLP stylometry
Message Routing	Use routing paths plausible for the institution	Routing rules fail	Graph-based deviation detection

Legitimate Network Behavior That Could Look Similar

Legitimate Scenario	Why It Looks Similar	False-Positive Risk
Legitimate large cross-border acquisition payment	Large amount, new beneficiary, urgent timing	Blocking legitimate M&A activity
Legitimate new correspondent relationship	New beneficiary BIC, routing through intermediary	Blocking legitimate business expansion
Emergency payment during holiday/weekend	Off-hours submission, urgent flags	Blocking legitimate emergency funding
Operator working from home/alternate location	Login from unusual IP, abnormal timing	Blocking legitimate remote work
Legitimate high-volume trading day	Rapid sequence of large payments	Blocking legitimate market activity
SWIFT network outage causing retries	Duplicate/near-duplicate messages	Blocking legitimate retry attempts
Legitimate restructuring with new beneficiaries	Sudden increase in new beneficiary setups	Blocking legitimate business changes
Legitimate corporate treasury optimization	Unusual routing path to reduce fees	Blocking legitimate cost optimization
Legitimate backup operator using credentials	Operator login from unusual workstation	Blocking legitimate backup/failover operations

D. SIMULATION INTELLIGENCE

Field	Value
Simulation Type	Hybrid (Agentic for phishing/voice + Algorithmic for message generation and transaction sequencing)

Synthetic Entities Required

Synthetic Entity	Description	Generation Method
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Synthetic Entity	Description	Generation Method
Synthetic Payment Network	Simulated SWIFT environment with participant BICs, routing logic, message processing, and settlement simulation	Algorithmic
Synthetic Financial Institutions	Sending institution (compromised), receiving institution, correspondent banks; each with BIC, transaction history, operator profiles	Algorithmic
Synthetic Payment Operators	Operator personas with credential access, historical login patterns, typical message volumes, historical transaction patterns	Algorithmic
Synthetic SWIFT Messages	MT/MX messages (MT103, MT202, MT760, MT199, MT999) with varying formats, narrative fields, regulatory references	Algorithmic + Generative (LLM for narrative)
Synthetic Mule Beneficiaries	Attacker-controlled accounts with shell company identities, BIC/account details, transaction histories	Algorithmic
Synthetic Transaction Sequences	Historical legitimate transactions (baseline) → Fraudulent message injection → Mule movement → Crypto layering → Cash-out	Algorithmic
Synthetic Routing Context	Correspondent bank paths; preferred vs. non-preferred routing; intermediary selection	Algorithmic
Synthetic Timing Patterns	Operator login times; payment submission times; settlement timing; delay-to-detection	Algorithmic
Synthetic Participant Relationships	Normal BIC relationships (trade partners, correspondent banks); abnormal BIC relationships (fraudulent); graph patterns	Graph Generation
Synthetic Network Graph	Institution-to-institution payment graph; beneficiary network; mule-cluster connections	Graph Generation
Synthetic Geographic Context	Originating jurisdiction; destination jurisdiction; correspondent jurisdictions; regulatory regime differences	Algorithmic
Synthetic State Transitions	Normal operation → Compromise → Fraudulent message → Processing → Settlement → Detection (or non-detection)	Sequence Generation
GenAI-Generated Behavior	Phishing emails targeting operators (LLM-generated); Voice clone audio for verification bypass (deepfake generation); AI-generated bank correspondence	Agentic + Generative
Attack Labels	Message forgery; SWIFT credential theft; Payment instruction injection; Beneficiary substitution; Cross-border fraud	
Variant Parameters	Message type (MT103 vs. MT202 vs. MT760); Fraudulent payment size; Correspondent routing path; Timing (off-hours vs. business hours); Jurisdiction	
Mutation Parameters	Message narrative variation; Beneficiary BIC variation; Timing randomization; Routing path variation; Amount distribution variation; Operator behavior variation	
Legitimate Counterpart	Legitimate urgent cross-border payments; Legitimate new business relationships; Legitimate weekend operations; Legitimate backup operator usage; Legitimate emergency payments	
Realistic Edge Cases	Partial compromise (some messages fraudulent, some legitimate); Suspicious but legitimate transaction (e.g., genuine acquisition); Multi-institution simultaneous attack; Failed attack (blocked by Payment Controls); Attack detected and reversed; Attack undetected until reconciliation	

E. EVIDENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
SWIFT payment message forgery used in Bangladesh Bank attack (\$81M stolen)	CSO Online / Industry reporting	CSO Online	2017	Industry report	VERIFIED
Attackers use AI to generate convincing bank correspondence and MT message templates	LinkedIn / Industry	Industry	2025	Industry analysis	REASONABLE INFERENCE

Claim	Source	Publisher	Date	Evidence Type	Confidence
SWIFT Customer Security Controls Framework v2026 references deepfakes as AI-based threat	SWIFT	SWIFT	2026	Primary source	VERIFIED
Deepfake voice confirmations can validate fraudulent SWIFT instructions	Industry analysis	Industry	2025	Industry analysis	REASONABLE INFERENCE
GenAI significantly lowers technical barrier for cybercriminals	Visa	Visa	2026	Primary source	VERIFIED
89% of FIs identify deepfakes and GenAI as supercharging payment scams	Datos Insights	Datos Insights	2026	Industry research	VERIFIED
"What once required deep technical skill can now be executed with a prompt"	Visa	Visa	2026	Primary source	VERIFIED
GenAI tools enabling more convincing phishing campaigns, impersonation scams, and deepfakes	Visa	Visa	2026	Primary source	VERIFIED
SWIFT provides unique view on payment anomalies at network level	SWIFT	SWIFT	-	Primary source	VERIFIED
Fraudulent SWIFT MT103 message injection via message queue demonstrated	Reverse Labs	Reverse Labs	2020	Security research	VERIFIED
SWIFT network covers 40,000+ possible payment routes; 86% direct or one intermediary	SWIFT	SWIFT	-	Primary source	VERIFIED

F. PRODUCT MAPPING

Generate (Red Team)

Component	Implementation
Agentic Generation	Phishing emails targeting SWIFT operators (LLM-generated); Voice clone audio for verification bypass; Bank correspondence for supporting documentation
Algorithmic Generation	SWIFT MT/MX message generation with varying formats; Transaction sequences (legitimate baseline → fraudulent injection → mule movement → cash-out); Routing context and timing patterns; Participant relationships and graph structures
Parameters Varied	Message type; Fraudulent payment amount; Correspondent routing path; Timing (off-hours vs. business hours); Jurisdiction; Message narrative variation; Beneficiary BIC variation; Operator behavior variation
Mutation Parameters	Message narrative variation; Beneficiary BIC variation; Timing randomization; Routing path variation; Amount distribution variation; Operator behavior variation

Defend (Blue Team)

Component	Implementation
What to Detect	Unusual operator login times/locations; Novel beneficiary BICs; Routing anomalies; AI-generated narrative patterns; Beneficiary graph anomalies; Rapid crypto conversion after receipt
Network Features/Signals	transaction_amount; beneficiary_bic_novelty; beneficiary_risk_score; operator_login_time_anomaly; submission_hour; is_weekend; routing_path_anomaly; narrative_ai_score; message_volume_velocity; beneficiary_graph_centrality; operator_session_duration_anomaly; regulatory_ref_presence
Detection Approach	Rules (time, amount, novelty); Supervised ML (binary classification); Anomaly Detection (Isolation Forest, Autoencoder); Sequence Modeling (LSTM/Transformer); Graph Neural Network; NLP (LLM-based detectors); Ensemble Risk Score
Network-Wide Intelligence	SWIFT Payment Controls; Cross-institution beneficiary graph; Cross-rail pattern detection; Consortium intelligence sharing

Mitigation

Risk Level	Action	Description
Low	Allow	Process payment; log for future pattern analysis
Medium	Enhanced Monitoring	Flag for Payment Controls; increase monitoring velocity; add to temporary watchlist
Medium-High	Hold + Verification	Hold payment for 4-24 hours; require callback to known institution number; require verbal confirmation of payment details
High	Manual Review	Escalate to compliance/security team; review all recent operator activity; check for other anomalous messages

Risk Level	Action	Description
Critical	Reject + Participant Escalation	Reject payment; revoke compromised credentials; notify SWIFT (security incident); initiate fraud investigation; coordinate with receiving institution to freeze funds (if possible)
Post-Detection	Forensic Investigation	Determine attack timeline; identify compromised credentials; assess all messages sent during window; update controls; share intelligence with consortium/network

Evaluation

Metric	Target	Notes
Precision	High (>90%)	Avoid business disruption
Recall	Very High (>95%)	Due to financial impact
F1 Score	>0.92	Balance precision and recall
False Positive Rate (FPR)	Very Low (<1%)	Avoid business disruption
Time-to-Detection (TTD)	Minutes (not hours/days)	Critical for preventing fund movement
Time-to-Mitigation (TTM)	< 1 hour	Critical for preventing settlement

PART 2 – NORMALIZATION DECISIONS

Candidate Classification Table

Candidate	Decision	Rationale
R-001: SWIFT Payment Message Forgery	KEEP	Genuinely rail-primary; SWIFT network's messaging and participant trust directly targeted; GenAI enhances phishing and deepfake verification (PARTIAL)
FASTCash (ISO 8583 Manipulation)	MOVE → Gateway/Processor	Payment switch is processor infrastructure; authorization decision is overridden; already covered as traditional malware
Fake Payment Gateway / Aggregator	MOVE → Merchant	Already covered in Merchant taxonomy (MCH-002, MCH-003); GenAI-enabled fake storefronts
Coordinated Payment Fraud Networks	MOVE → Cross-stage / Network-level	Spans multiple stages; requires network-wide detection; not a single Payment Rail attack
Routing Manipulation (Traditional)	DROP	Theoretical; no real-world evidence; GenAI is not load-bearing
Routing Manipulation (Agentic)	MOVE → Gateway/Processor (GP-005)	Already covered in Gateway/Processor taxonomy as AI Agent Routing Manipulation
Replay / Duplicate Transaction Attack	DROP	Already covered in Gateway/Processor; GenAI is incidental
Network-Level Fraud Control Evasion	MOVE → Cross-stage / Network-level	Distributed behavior across institutions; not a single Payment Rail attack
Participant Impersonation / Trust Abuse	MOVE → Authentication	Already covered in Authentication taxonomy (AUTH-001, AUTH-002, AUTH-003)
Cross-Border / Multi-Network Abuse	MOVE → Cross-stage / Network-level	Campaign-level attack exploiting regulatory gaps; not a single Payment Rail attack
Network Infrastructure Compromise	DROP	Generic cybersecurity; no payment-fraud consequence specific to rail

PART 3 – FINAL COVERAGE AUDIT

Payment Rail Attack Surface	Considered?	Covered By	Separate Family?	Reason
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Payment Rail Attack Surface	Considered?	Covered By	Separate Family?	Reason
Network message manipulation	✓	R-001	Yes	SWIFT message forgery
Transaction identifier manipulation	✓	MOVED → Gateway/Processor	No	Part of ISO 8583 manipulation (FASTCash)
Status/response manipulation	✓	MOVED → Gateway/Processor	No	FASTCash modifies response codes
Sequencing anomalies	✓	MOVED → Gateway/Processor	No	Part of replay/duplicate detection
Duplicate propagation	✓	DROPPED	No	GenAI not load-bearing
Replay	✓	DROPPED	No	GenAI not load-bearing
Network routing manipulation	✓	DROPPED (traditional)/ MOVED → GP-005 (agentic)	No	Theoretical; agentic covered in Gateway/Processor
Participant routing abuse	✓	DROPPED	No	Theoretical
Transaction-path manipulation	✓	DROPPED	No	Theoretical
Destination manipulation	✓	DROPPED	No	Theoretical
Participant identity abuse	✓	MOVED → Authentication	No	Credential theft primary
Participant impersonation	✓	MOVED → Authentication	No	Already covered in Authentication
Network trust abuse	✓	R-001	Yes	SWIFT trust model directly targeted
Unauthorized network access	✓	MOVED → Authentication/Employee	No	Generic cybersecurity
Network-level monitoring evasion	✓	MOVED → Cross-stage	No	Distributed behavior
Distributed transaction behavior	✓	MOVED → Cross-stage	No	Not a single Payment Rail attack
Threshold manipulation	✓	MOVED → Cross-stage	No	Distributed behavior
Cross-participant fraud	✓	MOVED → Cross-stage	No	Campaign-level
Coordinated network behavior	✓	MOVED → Cross-stage	No	Campaign-level
Cross-border/multi-network abuse	✓	MOVED → Cross-stage	No	Campaign-level
Network infrastructure manipulation	✓	DROPPED	No	Generic cybersecurity
Network configuration abuse	✓	DROPPED	No	Generic cybersecurity
Network administration abuse	✓	DROPPED	No	Generic cybersecurity
Adaptive GenAI network interaction	✓	R-001	Yes	LLM-generated messages; adaptive phishing
Agentic transaction behavior	✓	MOVED → GP-005	No	Already covered in Gateway/Processor

PART 4 – FINAL SUMMARY

Table 1 – FINAL PAYMENT RAIL ATTACK TAXONOMY

ID	Attack Family	Variants	GenAI Classification	Simulation Type	Primary Stage	Network Control	Reason for Distinctness
R-001	GenAI-Enhanced Payment Network Message Forgery	AI-generated SWIFT MT/MX messages; Hyper-personalized phishing; Deepfake voice verification bypass; AI-generated bank correspondence	PARTIAL	Hybrid	Payment Rail	Participant authentication; Message integrity; Operator trust	SWIFT network's messaging and participant trust directly targeted; GenAI enables hyper-personalized phishing at scale and deepfake voice verification –qualitative leap in credential theft and verification bypass

Table 2 – ATTACK × OBSERVABLE SIGNAL

Attack Family	Transaction	Network	Participant	Routing	Behavioral	Graph	Temporal	Geographic	Message/State
R-001	Amount deviation; Novel beneficiary BIC; Urgent flags; Narrative anomalies	Unusual message volume from participant; Unusual routing paths	Operator login time/location anomalies; Rapid message sequences	Unusual BIC combinations; Routing through non-preferred correspondents	Unnatural navigation; Superhuman speed	Beneficiary-mule connections; Rapid crypto conversion; Fraud ring clustering	Off-hours/holiday submissions; Rapid succession; Historical timing deviation	Unusual jurisdiction; No prior relationship; High-risk corridor	Format anomalies; AI-generated narrative; Missing/inconsistent regulatory references

Table 3 – ATTACK × SIMULATION

Attack Family	Synthetic Rail	Network	Participants	Transactions	Messages	Routing	Graph	Agentic/Algorithmic/Hybrid
R-001	SWIFT environment with BICs, routing, settlement	Correspondent banks; BIC relationships	Sending institution; Receiving institution; Operators; Mules	Historical baseline → Fraudulent injection → Mule movement → Crypto → Cash-out	MT103, MT202, MT760, MT199, MT999	Preferred vs. non-preferred; Intermediary selection	Beneficiary graph; Mule clusters; Institution relationships	Hybrid (Agentic: phishing/voice; Algorithmic: messages/routing)

Table 4 – TAXONOMY DECISIONS

Candidate	Decision	Reason
SWIFT Payment Message Forgery	KEEP	Genuinely rail-primary; SWIFT network's messaging and participant trust directly targeted
FASTCash	MOVE → Gateway/Processor	Payment switch is processor infrastructure; authorization decision overridden
Fake Gateway/Aggregator	MOVE → Merchant	Already covered in Merchant taxonomy
Coordinated Payment Fraud Networks	MOVE → Cross-stage	Spans multiple stages; not a single Payment Rail attack
Routing Manipulation (Traditional)	DROP	Theoretical; no real-world evidence; GenAI not load-bearing
Routing Manipulation (Agentic)	MOVE → Gateway/Processor (GP-005)	Already covered in Gateway/Processor
Replay / Duplicate Transaction	DROP	Already covered in Gateway/Processor; GenAI incidental
Network-Level Fraud Control Evasion	MOVE → Cross-stage	Distributed behavior; not a single Payment Rail attack
Participant Impersonation	MOVE → Authentication	Already covered in Authentication
Cross-Border / Multi-Network Abuse	MOVE → Cross-stage	Campaign-level; not a single Payment Rail attack
Network Infrastructure Compromise	DROP	Generic cybersecurity; no payment-fraud consequence

FINAL SUMMARY METRICS

Metric	Value
Total final attack families	1
Total variants	4 (AI-generated SWIFT MT/MX messages; Hyper-personalized phishing; Deepfake voice verification bypass; AI-generated bank correspondence)
Important candidates merged	None (only one family)

Metric	Value
Important candidates moved	FASTCash → Gateway/Processor; Fake Gateway → Merchant; Coordinated Networks → Cross-stage; Routing Manipulation (Agentic) → Gateway/Processor (GP-005); Participant Impersonation → Authentication; Control Evasion → Cross-stage; Cross-Border Abuse → Cross-stage
Important candidates dropped	Routing Manipulation (Traditional); Replay / Duplicate Transaction; Network Infrastructure Compromise
Theoretical/future threats	Routing manipulation in traditional payment rails (theoretical but insufficient evidence); AI-generated SWIFT message injection via API abuse (emerging but not yet verified); Agentic payment network interaction beyond current evidence
Remaining research gaps	Real-world frequency of GenAI-enhanced SWIFT message forgery; Specific detection evasion effectiveness against SWIFT Payment Controls; Impact of emerging AI architectures on network-level fraud detection; Effectiveness of cross-institution intelligence sharing against GenAI-enhanced attacks
Payment Rail control surfaces covered	Participant authentication; Message integrity; Payment operator trust; Settlement messaging; Routing validation
Important Payment Rail surfaces not represented as separate families	ISO 8583 manipulation (moved to Gateway/Processor); Routing manipulation (dropped: theoretical); Replay/duplicate (dropped: GenAI not load-bearing); Participant impersonation (moved to Authentication); Network infrastructure compromise (dropped: generic cybersecurity)

UNIFYING THEME

The Payment Rail taxonomy has a single unifying theme:

The Payment Rail: The Protected Core

Unlike previous lifecycle stages where GenAI creates **new attack families**, the Payment Rail's core remains fundamentally protected:

"The rail itself is the most protected layer. GenAI's impact is indirect—enhancing the attacks that enable rail compromise (phishing, deepfakes) rather than attacking the rail's core switching and routing functions."

Key Insights:

- The Rail is Not the Weakest Link:** Participant authentication, message integrity, and routing are heavily controlled. The rail's core trust model remains fundamentally intact against direct GenAI manipulation.
- GenAI Attacks "Around" the Rail:** GenAI's primary impact is on credential theft (phishing) and verification bypass (deepfakes)—the attacks that **enable** rail compromise, not attacks on the rail itself.
- One Family, Well-Defined:** The single surviving family (SWIFT message forgery) is PARTIAL because GenAI amplifies but does not create the attack. This reflects the reality that the rail is a **protected core** rather than a primary attack surface.
- Cross-Stage Attacks Dominate:** Most "network-level" fraud (coordinated networks, control evasion, cross-border abuse) belongs in the **Cross-stage/Network-level** category—they span multiple lifecycle stages and require network-wide visibility to detect, not just rail-level controls.
- Previous Stages Already Cover the Rest:** GenAI-enabled merchant fraud, gateway/processor attacks, authentication bypass, and acquirer manipulation are already captured in their respective lifecycle stages. The Payment Rail taxonomy should remain focused on what is genuinely rail-primary.

This concludes the Payment Rail Master Fraud Intelligence Document. The taxonomy is now consistent in structure and depth with the finalized Identity/KYC, Account Creation, Device/Session, Authentication, Payment Initiation, Merchant, Acquirer, and Gateway/Processor documents.